

0 math voodoo

$$\int u \, dv = uv - \int v \, du, \text{ partial fractions,}$$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n, |x-a| < \left(\lim_{n \rightarrow \infty} \left| \frac{f^{(n+1)}(a)}{f^{(n)}(a)} \right| \right)^{-1}, \sqrt{a^2 - (x+b)^2} \Rightarrow x+b = a \sin \theta$$

$$\frac{1}{1-x} = 1 + x + \dots + x^n + \dots, |x| < 1$$

$$e^x = 1 + \frac{x}{1!} + \dots + \frac{x^n}{n!} + \dots, x \in \mathbb{R}$$

1 Combinatorial Analysis

Generalised Principle of Counting

For r experiments, where the i -th experiment can result in any of n_i possible outcomes, we have $n_1 \cdot n_2 \cdot \dots \cdot n_r$ outcomes.

Permutations

- $n!$ arrangements of n distinct objects
- $\frac{n!}{n_1!n_2!\dots n_r!}$ arrangements of n objects of which n_1 are alike, n_2 are alike, ..., n_r are alike

Combinations

- $\binom{n}{r} = \frac{n!}{(n-r)!r!}$, for $r \leq n$
- $\binom{n}{r}$ subsets of size r from a set of size n .
- $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$, $1 \leq r \leq n$
- $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$
- $\binom{n}{n_1, n_2, \dots, n_r}$ divisions of n distinct objects into r distinct groups sized $n_1, n_2, \dots, n_r$ where $n_1 + n_2 + \dots + n_r = n$.
- For $n_1 + \dots + n_r = n$ and $n_i \geq 0$, $(x_1 + \dots + x_r)^n =$

$$\sum_{(n_1, \dots, n_r): n_1 + \dots + n_r = n} \binom{n}{n_1, \dots, n_r} x_1^{n_1} x_2^{n_2} \dots x_r^{n_r}$$

- Integer solutions for $n_1 + n_2 + \dots + n_r = n$, $n_i > 0$: $\binom{n+r-1}{r-1}$ distinct vectors $(n_1, n_2, \dots, n_r) \in \mathbb{N}^r$
- $\binom{n-1}{r-1}$ distinct vectors $(n_1, n_2, \dots, n_r) \in \mathbb{N}^{*r}$

2 Axioms of Probability

Event $E \subseteq$ Sample Space S

- Sample Space S** Set of all possible outcomes of an experiment
- Mutually Exclusive Events E, F** $\Leftrightarrow E \cap F = \emptyset$
- $E \subseteq F \Leftrightarrow \forall x \in E (x \in F)$
 $\Leftrightarrow E \cap F = E$

$$E = F \Leftrightarrow E \subseteq F \wedge F \subseteq E$$

Properties of $\cup$ and $\cap$:

Commutative. Associative. Distributive over the other.

DeMorgan's Laws

$$\left(\bigcup_{i=1}^n E_i \right)^c = \bigcap_{i=1}^n E_i^c \text{ and } \left(\bigcap_{i=1}^n E_i \right)^c = \bigcup_{i=1}^n E_i^c$$

Probability: Limiting Relative Frequency

For an experiment performed repeatedly many times under the same conditions, the probability of an event E happening is defined:

$$P(E) = \lim_{n \rightarrow \infty} \frac{n(E)}{n}$$

where $n(E)$ = frequency of E in n repetitions of the experiment

Kolmogorov's Axioms of Probability

For any event E , assume $P(E)$ is defined and satisfies axioms:

- $0 \leq P(E) \leq 1$
- $P(S) = 1$
- For any sequence of mutually exclusive events $E_1, E_2, \dots$,

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

$P(\cdot)$ satisfying the three axioms is known as a probability function.

Law of Large Numbers

For $X_1, X_2, \dots$ taken independently from a population, $\sum_{i=1}^n \frac{X_i}{n}$

almost surely (ie. with probability 1) converges to $E(X_i)$.

Probabilities!!!

- $P(E^c) = 1 - P(E)$
- $E \subseteq F \Rightarrow P(E) \leq P(F)$
- $\bigcup_{i=1}^n E_i = E_1 \cup E_1^c E_2 \cup E_1^c E_2^c E_3 \cup \dots$
 $\cup E_1^c \dots E_{n-1}^c E_n$
- Principle of Inclusion Exclusion (PIE)

$$\begin{aligned} P\left(\bigcup_{i=1}^n E_i\right) &= P(E_1 \cup E_2 \cup \dots \cup E_n) \\ &= \sum_{i=1}^n P(E_i) - \sum_{i_1 < i_2} P(E_{i_1} E_{i_2}) + \dots \\ &\quad + (-1)^{r+1} \sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} E_{i_2} \dots E_{i_r}) \\ &\quad + \dots + (-1)^{n+1} P(E_1 E_2 \dots E_n) \\ 5. \quad P\left(\bigcup_{i=1}^n E_i\right) &\leq \sum_{i=1}^n P(E_i) \\ &\geq \sum_{i=1}^n P(E_i) - \sum_{j < i} P(E_i E_j) \\ &\leq \sum_{i=1}^n P(E_i) - \sum_{j < i} P(E_i E_j) \\ &\quad + \sum_{k < j < i} P(E_i E_j E_k) \end{aligned}$$

Sample Space with Equally Likely Outcomes

$S = \{e_1, e_2, \dots, e_n\}$ where $P(\{e_1\}) = \dots = P(\{e_n\})$
 $P(\cdot)$ satisfies axioms $\Rightarrow \forall E \subseteq S, P(E) = \frac{|E|}{|S|}$

Probability as a Continuous Set Function

- $E_1, E_2, \dots$ is an increasing sequence if $E_1 \subset E_2 \subset \dots$
- Increasing Sequence $\{E_n, n \geq 1\} \Rightarrow$

$$\lim_{n \rightarrow \infty} P(E_n) = P\left(\lim_{n \rightarrow \infty} E_n\right) \equiv P\left(\bigcup_{i=1}^{\infty} E_i\right)$$

- $E_1, E_2, \dots$ is a decreasing sequence if $E_1 \supset E_2 \supset \dots$
- Decreasing Sequence $\{E_n, n \geq 1\} \Rightarrow$

$$\lim_{n \rightarrow \infty} P(E_n) = P\left(\lim_{n \rightarrow \infty} E_n\right) \equiv P\left(\bigcap_{i=1}^{\infty} E_i\right)$$

Probability as a Measure of Belief

Definition of probability as a measures of belief should still satisfy all of Kolmogorov's three axioms.

3 Conditional Probability and Independence

Conditional Probability

Probability E occurs given that F occurred is denoted $P(E|F)$

$$P(E|F) = \frac{P(EF)}{P(F)}, P(F) > 0$$

Conditional probability is a probability because it satisfies Kolmogorov's three axioms.

- $0 \leq P(E|F) \leq 1$

- $P(S|F) = 1$
- If E_i for $i = 1, 2, \dots$ are mutually exclusive events, then

$$P\left(\bigcup_{i=1}^{\infty} E_i | F\right) = \sum_{i=1}^{\infty} P(E_i | F)$$

Given that F occurred, occurrence of E is in EF . Only need to focus on F , the *reduced sample space*.

- $P(EF) = P(F)P(E|F)$
- $P(E_1 E_2 \dots E_n) = P(E_1)P(E_2|E_1) \dots P(E_n|E_1 \dots E_{n-1})$
- For $F_1, F_2, \dots, F_n$ partitioning S ,

$$P(E) = \sum_{i=1}^n P(EF_i) = \sum_{i=1}^n P(F_i)P(E|F_i)$$

- Baye's Theorem:** E occurred. Which F_j most likely led to E :

$$\begin{aligned} P(F_j|E) &= \frac{P(EF_j)}{P(E)} = \frac{P(F_j)P(E|F_j)}{P(E)} \\ &= \frac{P(F_j)P(E|F_j)}{\sum_{i=1}^n P(F_i)P(E|F_i)} \end{aligned}$$

- Odds of E** $= \frac{P(F)}{P(F^c)}$

Independent Events

- E is independent of $F \Leftrightarrow F$ is independent of E
 $\Leftrightarrow P(E|F) = P(E)$
 $\Leftrightarrow P(EF) = P(E)P(F)$
 $\Leftrightarrow E$ and F^c are independent
- n events $E_1, E_2, \dots, E_n$ are independent $\Leftrightarrow$ Any subset of the events are independent

4 Random Variables

Random variable maps sample space to a countable number of possible values.

$$X: S \mapsto \mathbb{R}$$

For some true value x_0 being measured by X , **mean squared error** $MSE = E[(X - x_0)^2] = Var(X - x_0) + [E(X - x_0)]^2 = Var(X) + (bias)^2$

- $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$

$$\begin{aligned} Var\left(\sum_{i=1}^n X_i\right) &= \sum_{i=1}^n Var(X_i) \\ &\quad + \sum_{i, j \in \{1..n\}, i \neq j} Cov(X_i, X_j) \end{aligned}$$

- For independent X_i ,

$$Var\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n Var(X_i)$$

4.1 Discrete Random Variables

PMF of a discrete random variable is $p(a) = P(X = a)$.
 $p(a) > 0$ for at most a countable number of values of a , and $\sum_{i=1}^{\infty} p(x_i) = 1$

CDF of a discrete random variable is $F(a) = P(X \leq a) = \sum_{all x \leq a} p(x)$ $F(a)$ is defined $\forall a \in \mathbb{R}$, and is a step function at every value of a for which $p(a) > 0$

$E(X)$ is the average of all possible values of X , weighted by the probabilities. $E(X) = \sum_x xp(x)$.

$$\text{For nonnegative integer-valued } X, E(X) = \sum_{i=1}^{\infty} P(X \geq i).$$

$$\text{For real-valued function } g(x), E[g(X)] = \sum_x g(x)p(x)$$

$$E(aX + b) = aE(X) + b, \text{ for some } a, b \in \mathbb{K}$$

$$\textbf{n-th Moment of } X: E(X^n) = \sum_x x^n p(x)$$

For random variable X with mean $\mu = E(X)$,

$$\begin{aligned} Var(X) &= E[(X - \mu)^2] \\ &= \sum_x (x - \mu)^2 p(x) \\ &= E(X^2) - [E(X)]^2 \end{aligned}$$

$$Var(aX + b) = a^2 Var(X), \text{ for some } a, b \in \mathbb{K}$$

Bernoulli X has probability of success p if

$$P(X = x) = \begin{cases} p & x = 1 \\ 1 - p & x = 0 \end{cases}$$

$X \sim \text{Bernoulli}(p)$, $E(X) = p$, $Var(X) = p(1 - p)$
Binomial X is the number of successes in n independent Bernoulli trials, each trial resulting in success with probability p .

$$\begin{aligned} P(X = k) &= \binom{n}{k} p^k (1 - p)^{n-k}, \forall k \geq 0 \\ &= \frac{(n - k + 1)p}{k(1 - p)} P(X = k - 1), \forall k \geq 1 \end{aligned}$$

$X \sim \text{Bin}(n, p)$, $E(X) = np$, $Var(X) = np(1 - p)$
 $X \sim \text{Bin}(n, p)$, $Y \sim \text{Bin}(m, p) \Rightarrow X + Y \sim \text{Bin}(n + m, p)$

Geometric X is the number of independent Bernoulli trials performed until a success is obtained (inclusive of the success), each trial resulting in success with probability p .

$$\begin{aligned} P(X = k) &= (1 - p)^{k-1} p, \forall k \geq 1 \\ X \sim \text{Geometric}(p), E(X) &= \frac{1}{p}, Var(X) = \frac{1-p}{p^2} \end{aligned}$$

Negative Binomial X is the number of independent Bernoulli trials performed until m successes are obtained (inclusive of the last success), each trial resulting in success with probability p . For $k \geq m$,

$$P(X = k) = \frac{(k-1)!}{(m-1)!(k-m)!} (1-p)^{k-m} p^m$$

$$X \sim NB(m, p), E(X) = n/p, Var(X) = \frac{m(1-p)}{p^2}$$

Hypergeometric X is the number of white balls selected in a sample of n balls without replacement from an urn of N balls of which m are white and $N - m$ are black. For $0 \leq i \leq n \wedge n - (N - m) \leq i \leq \min(n, m)$,

$$P(X = i) = \frac{\binom{m}{i} \binom{N-m}{n-i}}{\binom{N}{n}}$$

$X \sim \text{Hypergeometric}(N, m, n)$, $E(X) = \frac{nm}{N}$,
 $Var(X) = np(1 - p)(1 - \frac{n-1}{N-1})$ where $p = \frac{m}{N}$
 $n \ll N \Rightarrow X \sim \text{Bin}(n, \frac{m}{N})$, $Var(X) \approx np(1 - p)$

Poisson X has parameter λ if for some $\lambda > 0$, for $k \geq 0$,

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$X \sim \text{Poisson}(\lambda)$, $E(X) = \lambda$, $Var(X) = \lambda$
 $X \sim \text{Binomial}(n, p)$ $\wedge n$ is large $\wedge p$ is small $\Rightarrow Var(X) \approx np = E(X) \Rightarrow X \sim \text{Poisson}(np)$, even if trials are weakly dependent

Poisson X is the number of events that occur in any interval of length t , with the following assumptions:

- Probability that 1 event occurs in given interval of length h equals $\lambda h + o(h)$, where λ = occurrences per unit time
- Probability that ≥ 2 events occur in interval of length h equals

$o(h)$, for some $\lim_{h \rightarrow 0} \frac{o(h)}{h} = 0$
3. For $n, j_1, j_2, \dots, j_n \in \mathbb{N}$, a set of n nonoverlapping intervals and E_i the event that j_i of the events occur in the i -th interval, $E_1, E_2, \dots, E_n$ are independent

$X \sim \text{Poisson}(\lambda t)$. If $X \sim \text{Poisson}(\lambda_1), Y \sim \text{Poisson}(\lambda_2) \Rightarrow X + Y \sim \text{Poisson}(\lambda_1 + \lambda_2)$

4.2 Continuous Random Variables

X is a continuous random variable if $\exists f(x)$ such that

$$\forall x \in \mathbb{R}, f(x) \geq 0 \wedge P(X \in A) = \int_A f(x) dx$$

$f(x)$ is the PDF of X . CDF is defined $\forall a \in \mathbb{R}$,

$$F(a) = P(X \leq a) = \int_{-\infty}^a f(x) dx$$

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} P(X > x) dx \text{ for } X > 0.$$

For real-valued function $g(x)$, $E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$

$$Var(X) = E(X - \mu)^2 = E(X^2) - [E(X)]^2$$

Uniform X on the segment $[\alpha, \beta]$, PDF is:

$$f(x) = \begin{cases} \frac{1}{\beta - \alpha} & \alpha < x < \beta \\ 0 & \text{otherwise} \end{cases}$$

$X \sim U(\alpha, \beta)$, $E(X) = \frac{\alpha + \beta}{2}$, $Var(X) = \frac{(\beta - \alpha)^2}{12}$
Note: $(\alpha - \beta)U + \alpha \sim U(\alpha, \beta)$, $F(X) \sim U(0, 1)$,
 $F^{-1}(U)$ is a random variable with cdf $F(x)$

Normal has PDF on two parameters $\mu \in \mathbb{R}, \sigma > 0$:

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2}, x \in \mathbb{R}$$

$X \sim N(\mu, \sigma^2)$, $E(X) = \mu$, $Var(X) = \sigma^2$
 $Z \sim N(0, 1)$, CDF of Z denoted by $\Phi(z) = P(Z \leq z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} y^2} dy$, $\phi(x) = \frac{d}{dx} [\Phi(x)]$

Note: $X \sim N(\mu, \sigma^2) \Rightarrow \frac{X - \mu}{\sigma} \sim N(0, 1)$
 $Y \sim N(\mu, \sigma^2) \Rightarrow F_Y(a) \sim \Phi\left(\frac{a - \mu}{\sigma}\right)$

For independent $X_i \sim N(\mu_i, \sigma_i^2)$,

$$\sum_{i=1}^n X_i \sim N\left(\sum_{i=1}^n \mu_i, \sum_{i=1}^n \sigma_i^2\right)$$

Demoivre-Laplace Limit:

$$S_n \sim \text{Bin}(n, p) \Rightarrow \frac{S_n - np}{\sqrt{np(1-p)}} \sim Z, np(1-p) \geq 10$$

Gamma PDF on parameters $\alpha > 0, \lambda > 0$:

$$f(x) = \begin{cases} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$\forall x > 1, \Gamma(x) = \int_0^\infty u^{x-1} e^{-u} du = (x-1)\Gamma(x-1)$$

$X \sim \text{Gamma}(\alpha, \lambda)$, $E(X) = \frac{\alpha}{\lambda}$, $Var(X) = \frac{\alpha}{\lambda^2}$
For **independent random variables**:

$$\bullet X_i \sim \Gamma(\alpha_i, \lambda) \Rightarrow \sum_{i=1}^n X_i \sim \Gamma\left(\sum_{i=1}^n \alpha_i, \lambda\right)$$

$$\bullet Z_i \sim N(0, 1) \Rightarrow \sum_{i=1}^n Z_i^2 \sim \Gamma\left(\frac{n}{2}, \frac{1}{2}\right) = \chi_n^2$$

$$\bullet X_i \sim \text{Exp}(\lambda) \Rightarrow \sum_{i=1}^n X_i \sim \Gamma(n, \lambda)$$

Notably, the **Exponential** distribution is the case where $\alpha = 1$
 $F(x) = P\{X \leq x\} = 1 - e^{-\lambda x}$, $f(x) = \lambda e^{-\lambda x}$, $x \geq 0$

$X \sim \text{Exp}(\lambda)$, $E(X) = \frac{1}{\lambda}$, $Var(X) = \frac{1}{\lambda^2}$
The uniquely memoryless distribution (Discrete version-Geometric):
 $P(X > s + t | X > t) = P(X > s) \forall s, t > 0$

Beta PDF on parameters $a > 0, b > 0$:

$$f(u) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} u^{a-1} (1-u)^{b-1}, u \in [0, 1]$$

$$X \sim \text{Beta}(a, b), E(X) = \frac{a}{a+b},$$

$$Var(X) = \frac{ab}{(a+b)^2(a+b+1)}$$

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} = \int_0^1 x^{a-1} (1-x)^{b-1} dx$$

Note: $\text{Beta}(1, 1) = U(0, 1)$

Cauchy PDF on parameter $\theta \in \mathbb{R}$:

$$f(x) = \frac{1}{\pi} \frac{1}{1 + (x - \theta)^2}, x \in \mathbb{R}$$

$X \sim \text{Cauchy}(\theta)$, $\forall x \in \mathbb{N}$, $E(X^n)$ does not exist.

5 Random Multi-Variables

Joint Distribution Functions

Joint CDF $F(x, y) = P(X \leq x, Y \leq y)$

Marginal CDF $F_X(x) = \lim_{y \rightarrow \infty} F(x, y)$

For **discrete** random variables X and $Y, \forall (x, y) \in \mathbb{Z}^2$

Joint PMF $p(x, y) = P(X = x, Y = y)$

Marginal PMF

$$P(X = x) = \sum_{\{y: P(X=x, Y=y) > 0\}} P(X = x, Y = y)$$

Conditional Probability of X given that $Y = y, p_Y(y) > 0$

$$p_{X|Y}(x|y) = P(X = x | Y = y) = \frac{p(x, y)}{p_Y(y)}$$

$$p_X(x) = \sum_y p_{X|Y}(x|y) p_Y(y)$$

$$F_{X|Y}(x|y) = P(X \leq x | Y = y) = \sum_{a \leq x} p_{X|Y}(a|y)$$

For **jointly continuous** random variables $X, Y, \forall (x, y) \in \mathbb{R}^2$
 $\exists f(x, y) > 0, \forall C \subseteq \mathbb{R}^2$

$$P[(X, Y) \in C] = \iint_C f(x, y) dx dy$$

$$\text{Joint PDF } f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

$$\text{Marginal PDF } f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \frac{\partial}{\partial x} F(x, y)$$

Conditional Probability of X given $Y = y, f_Y(y) > 0$

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{X|Y}(x|y) f_Y(y) dy$$

$$F_{X|Y}(a|y) = P(X \leq a | Y = y) = \int_{-\infty}^a f_{X|Y}(x|y) dx$$

$$\text{Independence} \iff \forall A_1, A_2, \dots, A_n \subseteq \mathbb{R} \\ \iff P(X_1 \in A_1, \dots, X_n \in A_n)$$

$$= \prod_{i=1}^n P(X_i \in A_i)$$

$$\iff f_{X|Y}(x|y) = f_X(x), \forall x, y$$

For functions $U = g_1(X, Y), V = g_2(X, Y)$ of jointly continuous X, Y with $f_{XY}(x, y)$, if:

• Transformation is uniquely invertible $x = h_1(u, v), y = h_2(u, v)$

• g_1, g_2 have continuous partial derivatives

• $\forall x, y$ the Jacobian Determinant $J(x, y) \neq 0$

$$J(x, y) = \begin{vmatrix} \frac{\partial g_1}{\partial x} & \frac{\partial g_1}{\partial y} \\ \frac{\partial g_2}{\partial x} & \frac{\partial g_2}{\partial y} \end{vmatrix} = \left(\frac{\partial g_1}{\partial x}\right)\left(\frac{\partial g_2}{\partial y}\right) - \left(\frac{\partial g_1}{\partial y}\right)\left(\frac{\partial g_2}{\partial x}\right)$$

$$f_{UV}(u, v) = f_{XY}(x, y) \frac{1}{|J(x, y)|}$$

EXTENDABLE TO >2 VARIABLESAAAAAAAAAAAAA

6 Properties of Expectation

Expectation for Joint Variables

Suppose jointly distributed random variables $X_1, \dots, X_n$ and $Y = g(X_1, \dots, X_n)$,

$$E(Y) = E[g(X_1, \dots, X_n)]$$

Corollaries:

- Often can decompose X into a sum of n individual events. IMPT!
 $E(X_1 + X_2 + \dots + X_n) = E(X_1) + \dots + E(X_n)$
- For **independent** variables and fixed functions g and h ,
 $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$

Co's

Covariance of jointly distributed random variables X and Y is

$$\begin{aligned} Cov(X, Y) &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E(XY) - E(X)E(Y) \end{aligned}$$

$Cov(X, Y) < 0 \Rightarrow X$ and Y are inversely related

X and Y are independent $\Rightarrow Cov(X, Y) = 0$

X and Y are independent $\nRightarrow Cov(X, Y) = 0$

Results:

- $Cov(X, Y) = Cov(Y, X)$
- $Cov(X, X) = Var(X)$
- $Cov(aX, bY) = abCov(X, Y)$
- $Cov(X, Y + Z) = Cov(X, Y) + Cov(X, Z)$
- $Cov\left(\sum_{i=1}^n b_i X_i, \sum_{j=1}^m d_j Y_j\right) = \sum_{i=1}^n \sum_{j=1}^m b_i d_j Cov(X_i, Y_j)$

Correlation of X and Y is defined when

$\exists Var(X) \neq 0, Var(Y) \neq 0, Cov(X, Y)$

$$\rho = \rho(X, Y) = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

$-1 \leq \rho \leq 1$, with equality $\iff \exists a, b \in \mathbb{R}, P(Y = a + bX) = 1$

Correlation is the measure of *degree of linearity* between X and Y . Intuitively, the standardised covariance.

$\rho(X, Y) \rightarrow \pm 1 \Rightarrow$ high degree of linearity. $\rho(X, Y) = 0 \Rightarrow$ linearity is absent.

For indicator variables I_A, I_B , $Cov(I_A, I_B) = P(B)[P(A|B) - P(A)]$

Conditional Expectation of Y given $X = x$ is

$$\begin{aligned} E(Y|X = x) &= \sum_y p_{Y|X}(y|x) \\ &= \int_y y f_{Y|X}(y|x) dy \end{aligned}$$

Alternatively, it is the expectation on a reduced sample space consisting of outcomes for which $Y = y$

Conditional Variance of Y given X is

$$\begin{aligned} Var(Y|X = x) &= E[(Y - E(Y))^2 | X = x] \\ &= E(Y^2 | X = x) - [E(Y | X = x)]^2 \end{aligned}$$

$$\begin{aligned} Var(Y|X) &= E[(Y - E(Y))^2 | X] \\ &= E(Y^2 | X) - [E(Y | X)]^2 \end{aligned}$$

$E(Y|X), Var(Y|X)$ are functions of the random variable X , and hence also random variables.

Expectations!! Variances and Probabilities

• Law of Total Expectation

$$\begin{aligned} E(Y) &= E[E(Y|X)] \\ &= \sum_x E(Y|X = x) p_X(x) \end{aligned}$$

$$= \sum_x \left[\sum_y y p_{Y|X}(y|x) \right] p_X(x)$$

$$= \int_x E(Y|X = x) f_X(x) dx$$

$$= \int_x \left[\int_y y f_{Y|X}(y|x) dy \right] f_X(x) dx$$

- $Var(Y) = Var[E(Y|X)] + E[Var(Y|X)]$
- $P(E) = E[P(E|Y)]$

Moment Generating Functions

MGF of random variable X is $M(t) = E(e^{tX})$, if defined.

$$M(t) = \sum_x e^{tx} p(x) = \int_{-\infty}^{\infty} e^{tx} f(x) dx$$

$M^{(r)}$ continuous $\Rightarrow M^{(r)}(0) = \left. \frac{d^r}{dt^r} M(t) \right|_{t=0} = E(X^r)$

$X \sim \text{Bin}(n, p)$, $M(t) = [pe^t + (1-p)]^n$

$X \sim \text{Poisson}(n, p)$, $M(t) = e^{\lambda(e^t - 1)}$

$X \sim \text{Gamma}(\alpha, \lambda)$, $M(t) = \lambda^\alpha (\lambda - t)^{-\alpha}$

$X \sim N(\mu, \sigma^2)$, $M(t) = e^{\mu t + \frac{1}{2} \sigma^2 t^2}$

$Z = X + Y$, $M_Z(t) = M_X(t)M_Y(t)$ where mgf's exist

If mgf exists for $a < 0 < b$, it uniquely determines pdf

$Y = a + bX \Rightarrow M_Y(t) = e^{at} M_X(bt)$

$M_{XY}(s, t) = E(e^{sX + tY})$. $M_X(s) = M_{XY}(s, 0)$.

$M_{XY}(s, t) = M_X(s)M_Y(t)$.

X and Y are independent $\iff M_{XY}(s, t) = M_X(s)M_Y(t)$

7 Limit Theorems

• Markov's Inequality [$E(X)$ exists, and known], $\forall a > 0$

$$P(X \geq 0) = 1 \Rightarrow P(X \geq a) \leq \frac{E(X)}{a}$$

• Chebyshev's Inequality [Tighter, $\mu, Var(X)$ needed], $\forall k > 0$

$$P(|X - \mu| > k) \leq \frac{\sigma^2}{k^2}$$

• Chernoff Bounds $P(X \geq a) \leq e^{-ta} M_X(t), \forall t > 0$
 $P(X \leq a) \leq e^{-ta} M_X(t), \forall t < 0$

• Jensen's Inequality for convex $f: \forall x, f''(x) \geq 0$
 $E(f(X)) \geq f(E(X))$

• For i.i.d random variables $X_1, X_2, \dots, \forall \epsilon > 0$,
 $P\left(\left|\frac{X_1 + \dots + X_n}{n} - \mu\right| \geq \epsilon\right) \rightarrow 0$ as $n \rightarrow \infty$

• With probability 1, $\frac{X_1 + \dots + X_n}{n} \rightarrow \mu$ as $n \rightarrow \infty$

• For $Z_1, Z_2, \dots$ with cdf, mgf $F_{Z_n}(t), M_{Z_n}(t) \wedge \exists Z$ with $F_Z(t), M_Z(t)$, then where $F_Z(t)$ is continuous,
 $\forall t, M_{Z_n}(t) \rightarrow M_Z(t) \Rightarrow F_{Z_n}(t) \rightarrow F_Z(t)$

Central Limit Theorem

For i.i.d $X_1, X_2, \dots$ where $E(X_i) = \mu, Var(X_i) = \sigma^2$,
 $\frac{X_1 + \dots + X_n - n\mu}{\sigma\sqrt{n}} \sim N(0, 1)$ as $n \rightarrow \infty$